

David G. Wu

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☎ (860) 338-7222

in David Wu

Education

Aug. 2023 – Present

📖 **Ph.D. University of Texas at Austin**, Physics.

Advisor: Professor Aaron Zimmerman

Research Interests: Black hole perturbation theory and the calculation of quasi-normal modes in theories beyond-General Relativity

Aug. 2019 – May. 2023

📖 **B.A. Cornell University**, Physics

Advisor: Professor Saul Teukolsky and Professor Natasha Holmes

Research

Journal articles

- D. G. Wu, A. Hussain, and A. Zimmerman, “Computing spectral shifts for Johannsen-Psaltis black holes”, *Physical Review D* **113**, 104059 (2026)
- D. G. Wu, A. B. Heim, M. Sundstrom, C. Walsh, and N. G. Holmes, “Instructor interactions in traditional and nontraditional labs”, *Physical Review Physics Education Research* **18**, 010121 (2022)
- M. Sundstrom, D. G. Wu, C. Walsh, A. B. Heim, and N. G. Holmes, “Examining the effects of lab instruction and gender composition on intergroup interaction networks in introductory physics labs”, *Physical Review Physics Education Research* **18**, 010102 (2022)
- A. M. Phillips, M. Sundstrom, D. G. Wu, and N. G. Holmes, “Not engaging with problems in the lab: Students’ navigation of conflicting data and models”, *Physical Review Physics Education Research* **17**, 020112 (2021)

Presentations and Posters

- “Quasinormal modes of deformed black holes: parity and isospectrality breaking.” *Eastern Gravity Meeting*, May 2026, Ithaca, New York; Contributed talk
- “Quasinormal modes of deformed black holes: parity and isospectrality breaking.” *American Physical Society Global Physics Summit*, March 2026, Denver, Colorado; Contributed talk
- “Instructor interactions in traditional and nontraditional labs.” *American Physical Society April Meeting*, April 2022, New York, New York; Contributed talk
- “A spherical black hole gauge transformation for modeling rapidly spinning black holes with SpECTRE.” *American Physical Society April Meeting*, April 2022, New York, New York; Contributed talk co-author
- “Spherical Kerr-Schild coordinates in SpECTRE: a coordinate transform.” *Cornell Astronomy Research Experience for Undergraduates 2022 Symposium*, August 2022, Ithaca, New York; Poster

Computational Projects

- Contributor to SpECTRE, a highly scalable numerical relativity code: N. Deppe et al., “SpECTRE”, [10.5281/zenodo.15259042](https://doi.org/10.5281/zenodo.15259042)

Teaching and Mentoring

Spring 2026	Teaching Assistant for Waves and Optics Lab
Spring 2026	Mentor for UT Physics Directed Reading Program
Fall 2025	Teaching Assistant for Intro Electricity and Magnetism
Fall 2024-Spring 2025	Head Teaching Assistant for Intro Electricity and Magnetism Lab
Spring 2024	Teaching Assistant for Intro Electricity and Magnetism Lab
Fall 2023	Teaching Assistant for Intro Mechanics Lab
Fall 2021-Spring 2023	Society of Physic Students Peer Mentor
Spring 2021	Undergraduate Teaching Assistant for Intro Mechanics Lab

Skills

Coding	Proficient in Mathematica, Python, $\text{\LaTeX}$, Linux.
	Experience in C++, Matlab, Julia, R

Awards

Summer 2022	Roger and Mary Lou West Fellowship
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